Intellia SG — Reimagined Learning Flow Architecture

Inspired by “A Learner’s Journey” Narrative Framework

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1. EXPERIENCE TRANSFORMATION OVERVIEW

The original Place Value module follows a traditional:

INTRO → LEARN → PRACTICE → CHALLENGE → RESULTS

structure.

The new Intellia learning philosophy transforms the module into a cinematic, emotionally-driven, curiosity-first learning journey inspired by the visual flow shared in the reference image.

Instead of presenting lessons as educational tasks, the learner experiences the module as:

Wonder → Story → Simulate → Play → Reflect

This creates:

- Emotional engagement
- Memory retention through narrative
- Active experimentation
- Gamified mastery loops
- AI-assisted reflection
- Stronger long-term learning retention

The experience becomes:

"A learner’s journey from curiosity to mastery."

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2. NEW CORE LEARNING FLOW

INTELLIA 5-STAGE LEARNING JOURNEY

Stage	Experience Layer	Emotional Goal	Learning Goal
01	WONDER	Curiosity	Trigger exploration
02	STORY	Emotional connection	Contextual understanding

Stage	Experience Layer	Emotional Goal	Learning Goal
03	SIMULATE	Active experimentation	Concrete understanding
04	PLAY	Challenge & mastery	Reinforcement through gameplay
05	REFLECT	Self-expression	Deep retention

3. COMPLETE USER JOURNEY REDESIGN

3.1 FULL EXPERIENCE FLOW

```
WELCOME CINEMATIC
        ↓
AI GUIDE INTRODUCTION
        ↓
PHASE 01 — WONDER
(Mystery + Curiosity Hook)
        ↓
PHASE 02 — STORY
(Interactive Narrative)
        ↓
PHASE 03 — SIMULATE
(Hands-on Sandbox)
        ↓
PHASE 04 — PLAY
(Game World Challenge)
        ↓
PHASE 05 — REFLECT
(AI Conversation + Journal)
        ↓
CELEBRATION SCREEN
        ↓
KNOWLEDGE MAP UPDATE
        ↓
NEXT ADVENTURE UNLOCK
```

4. VISUAL EXPERIENCE SYSTEM

The entire UI must visually match the emotional richness and premium educational storytelling style shown in the reference image.

DESIGN LANGUAGE

Visual Direction

- Pixar-inspired educational aesthetic
- Rounded soft UI
- Rich gradients
- Warm emotional lighting
- Storybook composition
- Cinematic cards
- Interactive animated illustrations
- Soft floating particles
- Ambient motion

Mood

The platform should feel like:

- Disney + Duolingo + Khan Academy Kids + Nintendo

NOT:

- Traditional LMS
- Quiz platform
- Worksheet app

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5. STAGE-BY-STAGE REDESIGN

STAGE 01 — WONDER

PURPOSE

Create curiosity BEFORE teaching.

The learner should feel:

> “I want to know what happens next.”

EXPERIENCE FLOW

Opening Hook

The module begins with:

- Animated mystery scene

- Floating objects
- Soft cinematic music
- AI guide asks a curiosity question

Example:

"Why does 47 become 4 bundles and 7 tiny cubes?"

The answer is NOT immediately revealed.

INTERACTIONS

Curiosity Elements

- Tap hidden objects
- Reveal glowing clues
- Animated counting particles
- Interactive question marks
- Mini visual puzzles
- Prediction prompts

UI COMPONENTS

Component	Description
Mystery Card	Animated intro panel
Floating Hint Orb	Gives subtle clues
Curiosity Meter	Tracks exploration
AI Mascot Bubble	Speaks to learner
Discovery Glow Effects	Visual excitement

TECHNICAL REQUIREMENTS

Components

```
<WonderScene />
<CuriosityPrompt />
<DiscoveryAnimation />
<HintOrb />
<MysteryReveal />
```

Animation Requirements

- Parallax background
- Floating particles
- Light bloom
- Soft scale transitions
- Character eye tracking
- Hover interaction effects

Audio

- Ambient magical soundtrack
- Soft spark sounds
- Curiosity notification sounds

STAGE 02 — STORY

PURPOSE

Convert concepts into memorable emotional narratives.

The learner should feel:

> “I understand WHY this matters.”

EXPERIENCE FLOW

Instead of direct explanation:

A short interactive story introduces the concept.

Example:

"Wei Ming is building rocket fuel crates. Each large crate stores 10 energy cubes. How many crates and cubes are needed for 47 units?"

The child emotionally connects with:

- Character
- Mission
- Environment
- Goal

STORY SYSTEM

Narrative Structure

Part	Purpose
Character Intro	Emotional connection
Problem Scene	Context creation
Interactive Choice	Learner involvement
Visual Breakdown	Learning reveal
Mini Resolution	Reward loop

RICH CONTENT FEATURES

Story Elements

- Animated dialogue bubbles
- Character lip-sync
- Dynamic backgrounds
- Weather/day-night transitions
- Interactive objects
- Motion comics
- Camera zoom storytelling

TECHNICAL REQUIREMENTS

Components

```
<StoryScene />
<CharacterDialogue />
<InteractiveNarrative />
<AnimatedEnvironment />
<CinematicTransition />
```

Story Engine

```
const storyNode = {
  id: 'rocket_station_01',
  character: 'Wei Ming',
  dialogue: 'We need 47 energy cubes!',
  emotion: 'excited',
  choices: [...],
```

```
  rewards: [...]
}
```

STAGE 03 — SIMULATE

PURPOSE

Transform abstract concepts into physical understanding.

The learner should feel:

> "I discovered this myself."

EXPERIENCE FLOW

The child enters a sandbox laboratory.

Instead of solving questions:

- They experiment
- Drag objects
- Build numbers
- Break numbers apart
- Observe transformations

This is the core experiential layer.

SIMULATION ENVIRONMENTS

Environment Examples

Environment	Learning Purpose
Number Lab	Tens & ones building
Rocket Factory	Grouping into tens
Energy Grid	Skip counting
Cube Machine	Decomposition
Math Sandbox	Free exploration

ADVANCED INTERACTIONS

Physics-Based Learning

- Magnetic snapping
- Drag inertia
- Explosion decomposition
- Real-time counting
- Dynamic regrouping
- Animated number morphing

RICH VISUAL FEEDBACK

Feedback Effects

Event	Visual Response
Correct grouping	Energy burst
Tens formed	Rod fusion animation
Wrong placement	Gentle shake
Discovery moment	Glow pulse
Milestone complete	Mini celebration

TECHNICAL REQUIREMENTS

Simulation Components

```
<SimulationWorld />
<Base10Engine />
<PhysicsInteractionLayer />
<DynamicNumberRenderer />
<ExperimentPanel />
```

Technology Upgrades

- Framer Motion
- Canvas rendering
- Physics interactions
- Real-time animation state engine
- Drag-and-drop system
- Particle engine

STAGE 04 — PLAY

PURPOSE

Convert learning into mastery through challenge.

The learner should feel:

> “I can beat this level.”

EXPERIENCE FLOW

The learner enters:

INTELLIPLAY™

A game world where:

• Concepts become obstacles
• Questions become quests
• Scores become progression
• Mastery unlocks worlds

GAME STRUCTURE

Core Gameplay

Mechanic	Description
World Map	Multiple learning zones
XP System	Experience progression
Energy System	Continuous engagement
Coins & Rewards	Motivation loop
Boss Challenges	Concept mastery
Daily Quests	Retention system
Combo Streaks	Skill reinforcement

GAME MODES

Modes

Mode	Description
Adventure Mode	Story-driven progression
Arcade Mode	Fast challenge rounds
Time Rush	Speed learning
Builder Mode	Sandbox creativity
Team Challenge	Multiplayer collaboration

RICH GAME CONTENT

Gameplay Visuals

- Floating islands
- Animated enemies
- Achievement portals
- Reward explosions
- Level progression maps
- Dynamic environments
- Character customization
- Unlockable skins

TECHNICAL REQUIREMENTS

Components

```
<GameWorld />
<LevelEngine />
<XPTracker />
<RewardSystem />
<AchievementPanel />
<QuestManager />
```

Systems

```
const playerState = {
  level: 12,
  xp: 420,
  streak: 7,
  skins: [],
```

```
  unlockedWorlds: [],
  mastery: {
    placeValue: 0.84
  }
}
```

STAGE 05 — REFLECT

PURPOSE

Lock knowledge into long-term memory.

The learner should feel:

> "Now I truly understand this."

EXPERIENCE FLOW

Instead of ending with:

- Score only

The system asks:

- Explain in your own words
- Teach the AI mascot
- Record a voice reflection
- Draw your understanding

This creates:

- Metacognition
- Retention
- Personal ownership of learning

AI REFLECTION SYSTEM

Reflection Features

Feature	Purpose
AI Chat Buddy	Conversational reinforcement
Voice Reflection	Speaking learning aloud

Feature	Purpose
Journal Prompt	Writing memory encoding
Drawing Canvas	Visual understanding
Concept Summary	Personalized recap

EXAMPLE REFLECTION

AI: "Can you explain why 47 means 4 tens and 7 ones?"

Student: "Because 4 bundles are 40 and 7 cubes are extra ones."

AI: "Amazing! You just mastered place value."

TECHNICAL REQUIREMENTS

Components

```
<ReflectionScreen />
<AIConversation />
<VoiceRecorder />
<LearningJournal />
<ConceptSummary />
```

AI Integration

- Speech-to-text
- AI prompt engine
- Personalized encouragement
- Adaptive questioning
- Confidence analysis

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6. NEW INFORMATION ARCHITECTURE

OLD FLOW

```
Learn → Practice → Quiz
```

NEW FLOW

```
Emotion → Narrative → Experiment → Challenge → Reflection
```

This shifts the platform from:

Educational Software → Learning Experience Platform.

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7. CONTENT STRATEGY TRANSFORMATION

OLD CONTENT STYLE

- Static instructions
- Question-first learning
- Worksheet mentality

NEW CONTENT STYLE

- Narrative-first learning
- Exploration-based progression
- Emotionally engaging visuals
- Interactive storytelling
- Cinematic educational design

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8. REIMAGINED PRD STRUCTURE

PRODUCT EXPERIENCE PILLARS

Pillar	Description
Curiosity-Driven Learning	Questions before explanations
Story-Centered Retention	Narrative memory encoding
Simulation-Based Discovery	Learning by experimentation
Gameplay Reinforcement	Mastery through challenge
Reflective Intelligence	AI-powered retention

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9. REIMAGINED TECHNICAL ARCHITECTURE

HIGH-LEVEL ARCHITECTURE

```
<App>
 ├── <CinematicIntro />
 ├── <WonderEngine />
 ├── <StoryEngine />
 ├── <SimulationEngine />
 ├── <GameEngine />
 ├── <ReflectionAI />
 ├── <ProgressionSystem />
 ├── <RewardEngine />
 ├── <NarrationSystem />
 └── <KnowledgeMap />
```

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10. NEW CORE ENGINES

WONDER ENGINE

Controls:

- Curiosity prompts
- Discovery mechanics
- Mystery reveals
- Exploration interactions

STORY ENGINE

Controls:

- Narrative progression
- Character dialogue
- Emotion states
- Interactive branching

SIMULATION ENGINE

Controls:

- Physics interactions
- Sandbox mechanics

- Real-time visual learning
- Object transformations

GAME ENGINE

Controls:

- XP systems
- Levels
- Quests
- Progression
- Rewards
- Achievements

REFLECTION AI ENGINE

Controls:

- AI tutoring
- Journal prompts
- Voice interactions
- Personalized reinforcement

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11. DESIGN SYSTEM REIMAGINED

UI STYLE

Core Principles

- Soft depth
- Emotional color palette
- Child-safe contrast
- Motion-driven feedback
- Large touch interactions
- Rich animated storytelling

COLOR SYSTEM

Purpose	Color
Wonder	Electric Blue
Story	Warm Orange

Purpose	Color
Simulate	Scientific Purple
Play	Gaming Green
Reflect	Calm Indigo

MOTION SYSTEM

Motion Types

Motion	Purpose
Float	Magical feeling
Bounce	Positive feedback
Glow	Discovery moment
Shake	Gentle correction
Zoom	Focus attention
Burst	Celebration

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12. AI-POWERED LEARNING LAYER

INTELLIA LEARNING AI

AI Roles

Role	Purpose
Companion	Emotional support
Tutor	Guidance
Narrator	Storytelling
Challenger	Adaptive difficulty
Reflector	Deep learning reinforcement

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13. FUTURE SCALABILITY

This flow becomes the foundation for:

- Science simulations
- Coding adventures
- History narratives
- Geography exploration worlds
- AI-generated lessons
- Multiplayer learning quests
- Adaptive emotional learning systems

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14. FINAL EXPERIENCE VISION

The learner should never feel:

- "I am studying."

Instead they should feel:

- "I am exploring."
- "I am building."
- "I am discovering."
- "I am playing."
- "I am becoming smarter."

FINAL EXPERIENCE STATEMENT

INTELLIA

A Learner's Journey

From curiosity to mastery. Learning that stays.

Wonder. Story. Simulate. Play. Reflect.

5 Steps. One Journey. Infinite Potential.